

Simulating a Model - The Concentration of Digoxin Activity after Intravenous in Three Compartments

Introduction

Digoxin is a cardiac glycoside, a clinically significant drug with a low therapeutic index, meaning there is a very narrow window between an effective dose and a toxic one. This makes precise management critical, as even small deviations can lead to serious side effects. Understanding how the drug distributes through and is eliminated from the body is therefore an important step for clinicians when prescribing it. A pharmacokinetic (PK) model provides the simplest mathematical framework compatible with observed clinical data to describe how a drug moves through the body. Clinicians and researchers use these models to understand drug distribution, elimination, activity, and toxicity, and to predict concentrations and adjust dosages accordingly. The goal of the study by Narayan and Reddy was to compare two- and three-compartment open PK models against experimentally determined serum digoxin concentration data from human subjects. The researchers focused on the three-compartment model because it produced more accurate results and a better fit to clinical data. In this study, the three-compartment model is structured as a catenary model, where the drug flows linearly from the blood plasma to the intestinal fluid and finally into the cells, where elimination occurs. [1]

Methods

PK models represent the body as a series of interconnected volumes called compartments. These models generally follow one of two structures: the mammillary model, where peripheral compartments connect directly to a central compartment, and the catenary model, where compartments are arranged in a linear chain and the drug must pass through each one sequentially. For digoxin, the three compartments represent the blood plasma, the intestinal fluid, and the cells. Blood plasma is the vascular space where the drug is first introduced; intestinal fluid is the intermediate space through which the drug diffuses after leaving the blood; and the cells represent the intracellular space and deep tissues where the drug ultimately concentrates. This structure makes biological sense because digoxin must reach cellular targets, specifically the myocardium, to produce its therapeutic effect. The catenary model reflects the physical path the drug takes as it moves from blood through interstitial fluid and into the cellular environment.

The drug enters the system through the blood plasma compartment via an intravenous (IV) bolus dose (k_{01}). Unlike many standard models where elimination occurs from the central plasma compartment, elimination (k_{30}) here occurs at the third compartment (cells). This placement makes sense because the cellular space is where digoxin is most concentrated and active. Placing elimination there allows the model to more accurately reflect clinical

observations of how digoxin's concentration changes over time, and better captures the drug's high affinity for tissue rather than just its presence in the blood. [1]

The three-compartment catenary model for digoxin is described by three ordinary differential equations that represent the rate of change of the drug amount in each compartment over time.

Blood Plasma Compartment (x_1):

$$\frac{dx_1(t)}{dt} = -k_{12}x_1 + k_{21}x_2;$$

- dx_1/dt : The rate of change of the digoxin amount in the blood plasma
- $-k_{12}x_1$: This represents the drug leaving the plasma and diffusing into the intestinal fluid. Biologically, this is the first step of distribution after an IV injection
- $+k_{21}x_2$: This represents the drug returning to the plasma from the intestinal fluid, capturing the reversible nature of the distribution between the two spaces

Intestinal Fluid Compartment (x_2):

$$\frac{dx_2(t)}{dt} = k_{12}x_1 - (k_{21} + k_{23})x_2 + k_{32}x_3;$$

- dx_2/dt : The rate of change of the digoxin amount in the intermediate intestinal fluid
- $k_{12}x_1$: The rate at which the drug enters from the blood plasma
- $-k_{21}x_2$: The rate at which the drug leaves the intestinal fluid to return to the plasma
- $-k_{23}x_2$: This represents the drug moving forward from the intestinal fluid into the cellular compartment
- $+k_{32}x_3$: This represents the drug returning from the cells back into the intestinal fluid

Cells Compartment (x_3):

$$\frac{dx_3(t)}{dt} = k_{23}x_2 - (k_{32} + k_{20})x_3;$$

- dx_3/dt : The rate of change of the digoxin amount in the deep cellular tissues
- $k_{23}x_2$: The rate at which the drug enters the cells from the intestinal fluid
- $-k_{32}x_3$: The rate at which the drug leaves the cells to return to the intestinal fluid.
- $-k_{20}x_3$: This represents the elimination (clearance) of digoxin from the body. Biologically, this term accounts for the drug being metabolized or excreted directly from the cellular/tissue level rather than the plasma

For the IV bolus dose, the initial conditions at time zero ($t=0$) are: $x_1(0) = k_{01}$, which represents the total initial dose administered. Biologically, because the drug is injected directly into the vein (bolus), the entire amount is assumed to be present in the Blood Plasma (x_1) instantaneously before any distribution occurs, $x_2(0) = 0$, and $x_3(0) = 0$, where no drug is present in the intestinal fluid or cells at the exact moment of injection.

The rate constants (k_{ij}), also known as the transfer coefficients, relate the amount of drug being transferred per unit of time from one compartment to another:

- k_{12} and k_{21} : govern the distribution exchange between plasma and intestinal fluid
- k_{23} and k_{32} : Govern the distribution exchange between intestinal fluid and the cells
- k_{30} : represents the irreversible elimination of the drug from the system

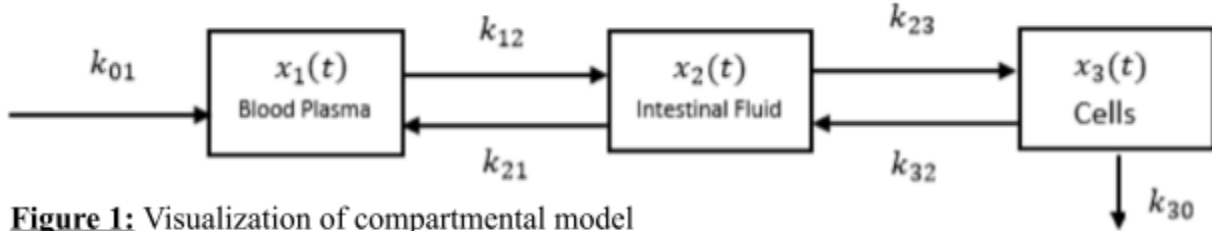


Figure 1: Visualization of compartmental model

MATLAB Implementation

The ODE system is written as a local function at the bottom of the script that takes **time**, the state vector, and **theta** as inputs. **theta** stores all five rate constants, which are unpacked inside the function and used to define the three differential equations. The anonymous function handle **odeFunc** wraps this into the two-argument format **ode15s** requires. **ode15s** was chosen because the system is stiff since plasma concentration drops rapidly in the first hour while elimination plays out over 72 hours, and **ode15s** handles this large difference in timescales more efficiently than a non-stiff solver. Time is kept in hours throughout because the paper's rate constants are in hr^{-1} , so using minutes would make the drug appear to disappear roughly 60 times faster than it actually does.

Observability describes how well a system's internal states can be determined from its measurable outputs. This is important for this experiment because it determines whether drug concentrations in the inaccessible compartments can be calculated from plasma data alone. Blood plasma (x_1) is the only practical measurement site, it is easily accessible, minimally invasive, and is the established clinical standard for monitoring low therapeutic index drugs like digoxin. Directly measuring intestinal fluid or cellular concentrations would require invasive procedures that are not feasible in a standard clinical setting. To verify this, the observability matrix $\mathbf{O} = [\mathbf{C}; \mathbf{C}\mathbf{A}; \mathbf{C}\mathbf{A}^2]$ was constructed using the built-in MATLAB function **obsv(A, C)**, where $\mathbf{A}$ is the system matrix and $\mathbf{C} = [1, 0, 0]$ selects the plasma compartment as the output. The rank of $\mathbf{O}$ was computed using **rank(O)** and compared against the number of states using **size(A,1)**. As shown in **Appendix B**, the observability matrix has a rank of 3 out of 3, with zero

hidden modes, confirming that the system is fully observable from blood plasma alone. This means the concentrations in the intestinal fluid and cell compartments can be mathematically reconstructed entirely from plasma measurements, without ever directly sampling those compartments. The controllability analysis, also shown in **Appendix B**, confirms a rank of 3 out of 3 with zero hidden modes, meaning all three compartment states can be driven from the plasma input, which is consistent with the IV bolus entering through x_1 .

Sampling Schedule

The sampling schedule follows a dense-then-sparse design to capture both phases of digoxin kinetics. In the first hour after the IV bolus, plasma concentration drops steeply as the drug rapidly distributes from blood into tissues. Missing this phase would make it impossible to estimate the distribution rate constants k_{12} and k_{21} . Dense sampling early on ensures these fast dynamics are captured accurately. As the drug approaches distributional equilibrium and enters the slower terminal elimination phase, the rate of change slows enough that sampling every few hours is sufficient to characterize the curve. This design mirrors the original study's Table 3.2, which samples as early as 2.4 minutes post-dose and extends to nearly 72 hours, and follows the general principle that sampling frequency should match the rate of change of the signal being measured.

Noise Model

Clinical monitoring of digoxin serum levels is primarily performed using a Fluorescence Polarization Immunoassay (FPIA), which is subject to analytical variability and a detection threshold of approximately 0.2 ng/mL. Log-normal noise was chosen over Gaussian noise because it prevents negative concentration values, which are physically impossible, and better reflects the proportional error behavior typical of laboratory assays. Since blood plasma is the only compartment that is directly measured, noise is applied exclusively to x_1 . The intestinal fluid and cell states are internal model states that are never directly sampled and therefore do not have measurement noise applied to them. In practice, a CV of 10% for a true plasma concentration of 2 ng/mL gives a standard deviation of 0.2 ng/mL ($SD = 2 \text{ ng/mL} \times 0.10 = 0.2 \text{ ng/mL}$), resulting in an expected measured range of approximately 1.6 to 2.4 ng/mL within a 95% confidence interval. [2]

Results

Figure 2 shows the true ODE solution for all three compartments with no noise applied. Plasma concentration (blue) peaks immediately at injection and declines rapidly as the drug distributes into tissues. Intestinal fluid (red) rises quickly then falls as it acts as an intermediate compartment, and the cells (green) accumulate the drug most slowly, peak latest around 5–6 hours, and maintain the highest concentration longest because k_{30} , the elimination rate, is the smallest rate constant in the model. This is consistent with digoxin's known high tissue affinity, particularly in the myocardium. **Figure 3** isolates the plasma compartment and overlays the

simulated noisy measurements, representing what would actually be collected in a clinical experiment through blood draws at each sample time. The X markers scatter around the true curve with the expected 10% variability, and the dense-then-sparse sampling pattern is clearly visible, many closely spaced measurements in the first hour, followed by increasingly spread timepoints through 72 hours. Both figures use a semi-logarithmic scale and arbitrary units. **Figure 2** shows the full internal dynamics of all three compartments, while **Figure 3** isolates the plasma state to show what would actually be measured in the experiment. The paper's figures (3.1–3.5) use a linear scale and clinical units (ng/mL) to validate calculated versus observed plasma concentration only. Despite these differences, both represent the same underlying distribution and elimination behavior for Subject T.F. The model's accuracy is further supported by RMSE values below 0.5 for all subjects, meaning the calculated concentrations closely match the clinical data.

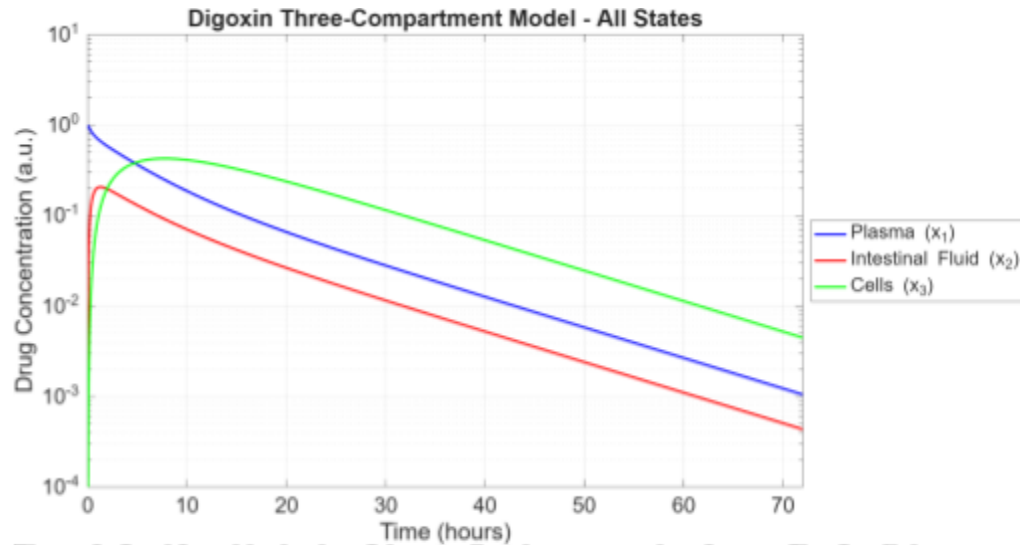


Figure 2: Semi-logarithmic plot of the true digoxin concentration-time profiles for all three compartments (Blood Plasma x_1 , Intestinal Fluid x_2 , Cells x_3) over 72 hours following an IV bolus dose, as predicted by the three-compartment catenary model for Subject T.F.

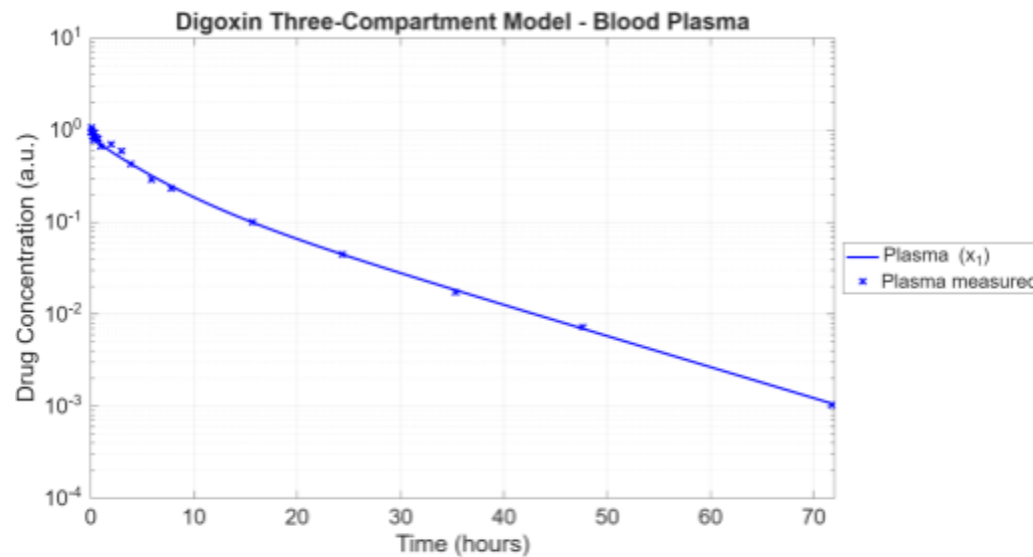


Figure 3: Semi-logarithmic plot of the true blood plasma concentration-time profile (x_1) with simulated noisy measurements ($\times$) representing blood draws at each sample time. Log-normal noise (CV = 10%) is applied to reflect clinical immunoassay variability.

Conclusion

The 3-compartment catenary model successfully characterized digoxin kinetics. A full observability rank of 3 mathematically proves that the hidden states (x_2 , x_3) are reconstructible from plasma measurements (Topic 5). Success is further validated by a low RMSE (< 0.5) and concentration-time curves that accurately mirror physiological distribution and elimination

phases. This project applies compartmental ODE modeling (Topic 4) and linear system theory (Topic 2) to a practical biomedical problem (Topic 1): predicting serum concentrations for managing drugs with narrow therapeutic windows. While the model effectively uses mass-action dynamics (Topic 6), limitations include significant inter-subject variability in rate constants, a reliance on normalized dosing, and the assumption of linearity, which may fail at toxic concentrations. Finally, the choice of log-normal noise keeps simulated measurements positive and reflects proportional measurement error.

Reference:

- [1] K. L. Narayan and V. S. R. K. Reddy, "The Concentration of Digoxin Activity after Intravenous in Three Compartments Mathematical Model," *Int. J. Recent Technol. Eng. (IJRTE)*, vol. 8, no. 2S11, Sept. 2019 DOI: [10.35940/ijrte.B1460.0982S1119](https://doi.org/10.35940/ijrte.B1460.0982S1119)
- [2] A. E. Lim, J. R. Tate, D. Clarke, R. L. Norris, R. G. Morris, and J. H. Martin, "Clinical consequences of a miscalibrated digoxin immunoassay," *Ther. Drug Monit.*, vol. 36, no. 5, pp. 603–610, Oct. 2014, doi: 10.1097/FTD.0000000000000118

Appendix A:

8	%% Parameters
9	
10	% Rate constants (hr⁻¹)
11	k = zeros(5,1);
12	k(1) = 0.5712; % k12: Plasma --> Intestinal Fluid
13	k(2) = 1.1934; % k21: Intestinal Fluid --> Plasma
14	k(3) = 0.6654; % k23: Intestinal Fluid --> Cells
15	k(4) = 0.0389; % k32: Cells --> Intestinal Fluid
16	k(5) = 0.1035; % k30: Cells --> Elimination
17	
18	theta = k;
19	
20	%% Initial Conditions
21	
22	% IV bolus of 1 unit into plasma at t = 0
23	qi = [1; 0; 0];
24	

zeros(n,1): creating a zero-filled vector before element-by-element assignment

```

133 %% ODE Function
134
135 function dydt = digoxin_ode(~, y, theta)
136
137 % Parameters
138 k12 = theta(1);
139 k21 = theta(2);
140 k23 = theta(3);
141 k32 = theta(4);
142 k30 = theta(5);
143
144 % States
145 x1 = y(1);
146 x2 = y(2);
147 x3 = y(3);
148
149 dx1 = -k12*x1 + k21*x2;
150 dx2 = k12*x1 - (k21 + k23)*x2 + k32*x3;
151 dx3 = k23*x2 - (k32 + k30)*x3;
152
153 dydt = [dx1; dx2; dx3];
154
155 end

```

```

25 %% Call ODE Function and Solve
26
27 odeFunc = @(t,y) digoxin_ode(t, y, theta);
28
29 tspan = [0 72]; % hours
30 [t, y] = ode15s(odeFunc, tspan, qi);
31
32 % Sample times (hr) - dense early to capture distribution phase, sparse later
33 t_sample = [0.033, 0.067, 0.1, 0.133, 0.167, 0.233, 0.3, 0.367, ...
34             0.5, 0.75, 1, 2, 3, 3.9, 5.9, 7.8, 15.7, 24.4, 35.3, 47.6, 71.7];
35
36 [~, ys] = ode15s(odeFunc, t_sample, qi);
37

```

```

38  %% Controllability and Observability Analysis
39
40  % State matrix A and input/output matrices
41  A = [-k(1),      k(2),      0;
42       k(1),  -(k(2)+k(3)),  k(4);
43       0,      k(3),  -(k(4)+k(5))];
44
45  B = [1; 0; 0];    % IV bolus enters x1 (Plasma)
46  C = [1, 0, 0];    % observe x1 (Plasma) only
47
48  % Controllability
49  CO      = ctrb(A, B);
50  rank_CO = rank(CO);
51  n       = size(A, 1);
52  hidden_CO = n - rank_CO;
53
54  fprintf('Controllability Matrix:\n'); disp(CO)
55  fprintf('Rank:          %d / %d\n', rank_CO, n)
56  fprintf('Hidden modes: %d\n', hidden_CO)
57  if hidden_CO == 0
58      fprintf('System is FULLY CONTROLLABLE from Blood Plasma.\n')
59  else
60      fprintf('System is NOT fully controllable. Hidden modes: %d\n', hidden_CO)
61      fprintf('Uncontrollable modes (null space):\n'); disp(null(CO'))
62  end
63

```

```

64  % Observability
65  O      = obsv(A, C);
66  rank_O = rank(O);
67  hidden_O = n - rank_O;
68
69  fprintf('Observability Matrix:\n'); disp(O)
70  fprintf('Rank:          %d / %d\n', rank_O, n)
71  fprintf('Hidden modes: %d\n', hidden_O)
72  if hidden_O == 0
73      fprintf('System is FULLY OBSERVABLE from Blood Plasma.\n')
74  else
75      fprintf('System is NOT fully observable. Hidden modes: %d\n', hidden_O)
76      fprintf('Unobservable modes (null space):\n'); disp(null(O))
77  end
78

```

obsv(A, C): In class, the observability matrix was built manually as $[C; C \cdot A; C \cdot A^2 \cdot A]$. In this script, the MATLAB built-in `obsv(A, C)` constructs the same matrix automatically for a system of any size.

size(A,1): returns the number of rows of matrix A (i.e., the number of states, $n=3$). This is used to compare against `rank(O)` without hard-coding the number 3, making the check generalize to any system size.


```

79      %% Simulate Noisy Measurements
80      % Log-normal noise applied to x1 (Plasma) only - the sole measurement site
81      % CV = 10% for digoxin immunoassay
82
83      sig_plasma = log(1.10);
84
85      rng(42);
86      x1_noisy = random('Lognormal', log(ys(:,1)), sig_plasma);
87

```

random('Lognormal', log(ys(:,1)), sig_plasma): log(ys(:,1)) is passed rather than ys(:,1) directly because the lognormal is parameterized in log-space

```

88      %% Figure 1: All Three States - no noise
89
90      colors      = {'b', 'r', 'g'};
91      labels_cont = {'Plasma (x_1)', 'Intestinal Fluid (x_2)', 'Cells (x_3)'};
92
93      figure(1)
94      hold on
95
96      for i = 1:3
97          plot(t, y(:,i), '-', 'Color', colors{i}, 'LineWidth', 2);
98      end
99
100     set(gca, 'YScale', 'log');
101     set(gca, 'FontSize', 20);
102     xlabel('Time (hours)')
103     ylabel('Drug Concentration (a.u.)')
104     title('Digoxin Three-Compartment Model - All States')
105     legend(labels_cont, 'Location', 'eastoutside')
106     xlim([0 72])
107     ylim([1e-4 10])
108     grid on
109     box on
110     hold off
111

```

```

112     %% Figure 2: Blood Plasma Only - noisy
113
114     figure(2)
115     hold on
116
117     plot(t, y(:,1), 'b-', 'LineWidth', 2, 'DisplayName', 'Plasma (x_1)');
118     plot(t_sample, x1_noisy, 'bx', 'LineWidth', 2, 'MarkerSize', 8, ...
119          'DisplayName', 'Plasma measured');
120
121     set(gca, 'YScale', 'log');
122     set(gca, 'FontSize', 20);
123     xlabel('Time (hours)')
124     ylabel('Drug Concentration (a.u.)')
125     title('Digoxin Three-Compartment Model - Blood Plasma')
126     legend('Location', 'eastoutside')
127     xlim([0 72])
128     ylim([1e-4 10])
129     grid on
130     box on
131     hold off
132

```

set(gca, 'YScale', 'log'): In class, semilogy was used to produce a log-scale y-axis. Here, set(gca, 'YScale', 'log') achieves the same result but is applied after all plot calls are complete.

Appendix B:

Outputs:

Controllability Matrix:

```

1.0000 -0.5712  1.0079
    0   0.5712 -1.3880
    0    0   0.3801

```

Rank: 3 / 3

Hidden modes: 0

System is FULLY CONTROLLABLE from Blood Plasma.

Observability Matrix:

```

1.0000    0    0
-0.5712  1.1934    0
1.0079 -2.9000  0.0464

```

Rank: 3 / 3

Hidden modes: 0

System is FULLY OBSERVABLE from Blood Plasma.